Generalized Fiducial Inference for Hybrid Discrete-Continuous Count Data

Kendall Thomas
Doctoral Candidate, Statistics and Operations Research
Joint work with Dr. Jan Hannig



The University
of North Carolina
at Chapel Hill

2026 AISC Conference
Invited Session 1: Bayesian and Fiducial Inference with Applications

Motivation

- **Quantile cube**: time spent in 5 × 5 × 4 = 100 velocity × acceleration × angle bins per athlete–match session

- Counts range from under 100 to several thousand deciseconds:
  - NB is exact but costly
  - a Normal approximation fails where the NB shape (mean × dispersion) is small

- **Goal:** uncertainty for 500 regression and dispersion parameters, without specifying a prior

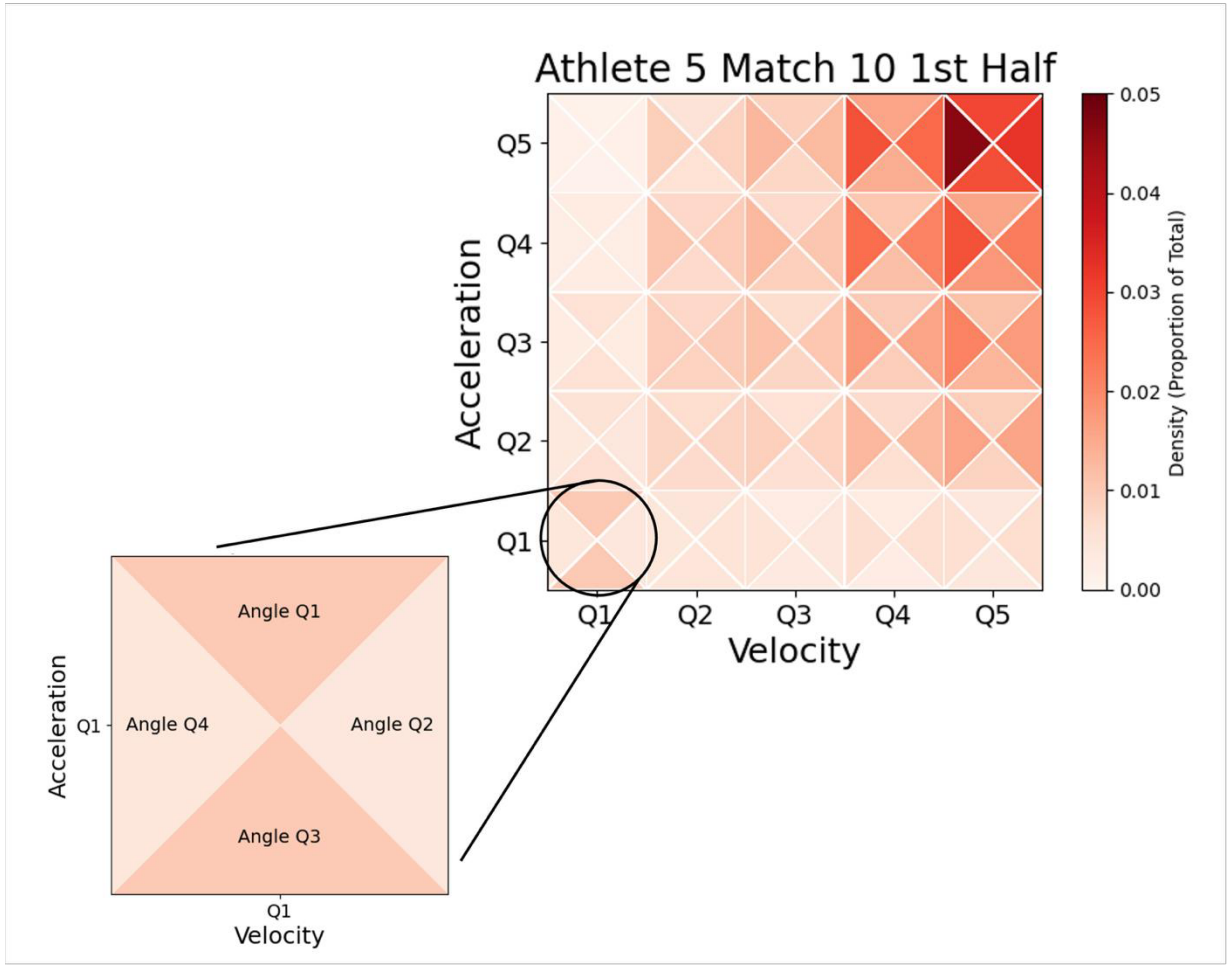
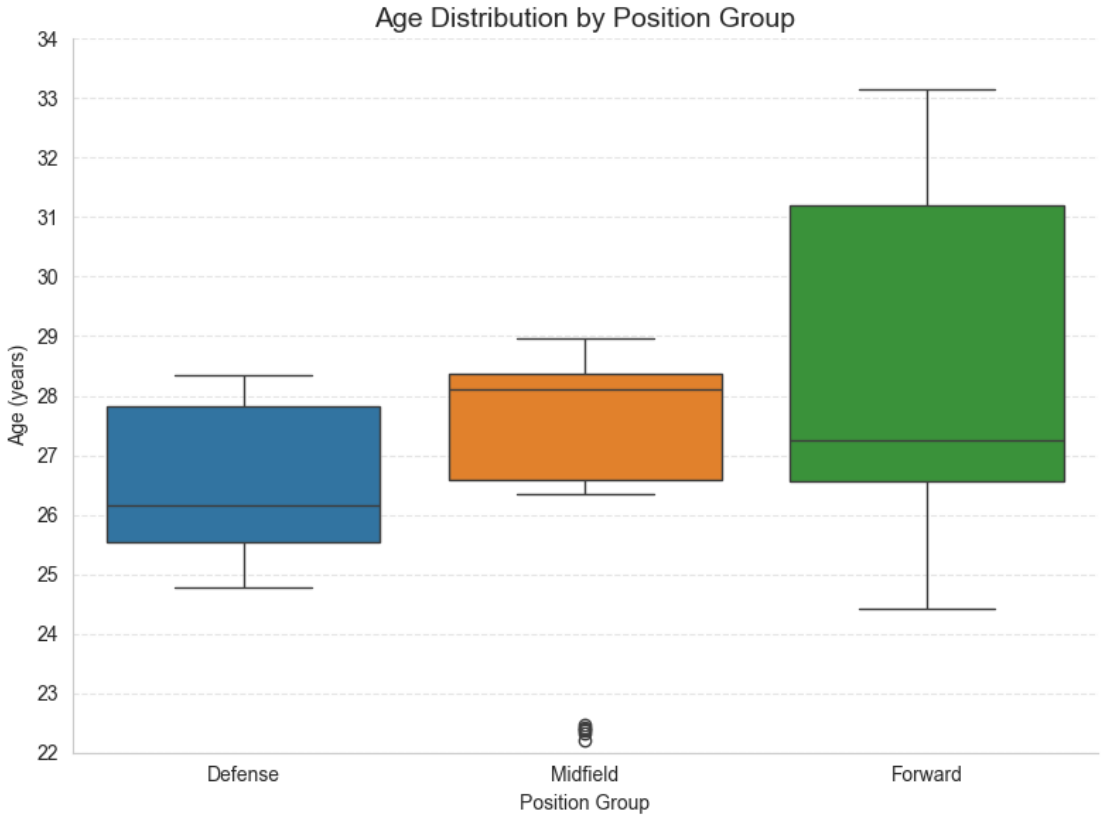


Fig. from [1] Thomas & Hannig (2026), Journal of Quantitative Analysis in Sports

The Data

- GPS tracking data from one season for a professional women's soccer team (IRB 25-1234)
  - Sampling at 100 Hz with access at 10 Hz: timestamp, longitude, latitude
- Retained athlete-match sessions with ≥ 60 minutes played in ≥ 3 matches
  - Total: 17 athletes, 26 matches → 216 athlete-match sessions
- Covariates:
  - standardized age (mean 27.3 yrs, SD 2.1; range 22.2 – 33.1)
  - position (6 defenders, 4 midfielders, 7 forwards)



The Model

- **Negative Binomial (NB) shape** $r_{ij} = \mu_{ij}\nu_j$ (mean × dispersion) decides how close count i in bin j is to Normal, not the count itself

- **Our Approach**: assign each count by a 95% lower bound $\hat{r}_{ij}^{L}$ on its shape, cutoff $\omega = 10$

$$y_{ij} \sim \begin{cases} \mathrm{NB}\left(\alpha_{ij}t_i, \dfrac{\nu_j}{1+\nu_j}\right) & \text{if } \hat{r}_{ij}^{L} < \omega, \\ \mathcal{N}\left(\mu_{ij} = \dfrac{\alpha_{ij}t_i}{\nu_j}, \sigma_{ij}^2 = \dfrac{\alpha_{ij}t_i(\nu_j+1)}{\nu_j^2}\right) & \text{if } \hat{r}_{ij}^{L} \geq \omega \end{cases}$$

where $\alpha_{ij} = exp({X_i}^{\mathsf{T}}\beta_j)$ and $\boldsymbol{\theta} = ({\beta_1}^{\mathsf{T}}, \ldots, {\beta_{100}}^{\mathsf{T}}, \nu_1, \ldots, \nu_{100})^{\mathsf{T}}$

The Generalized Fiducial Density

- Generalized Fiducial Inference (GFI) expresses the data through a structural equation $Y = G(U,\ \theta)$ involving the parameters $\boldsymbol{\theta}$ and auxiliary variables $\mathbf{U}$ with known distribution[2, 3]

Theorem 2: The joint generalized fiducial density takes the form

$$r(\boldsymbol{\theta} \mid \mathbf{y}) \propto L(\mathbf{y} \mid \boldsymbol{\theta})\, J(\boldsymbol{\theta}, \mathbf{y_C})$$

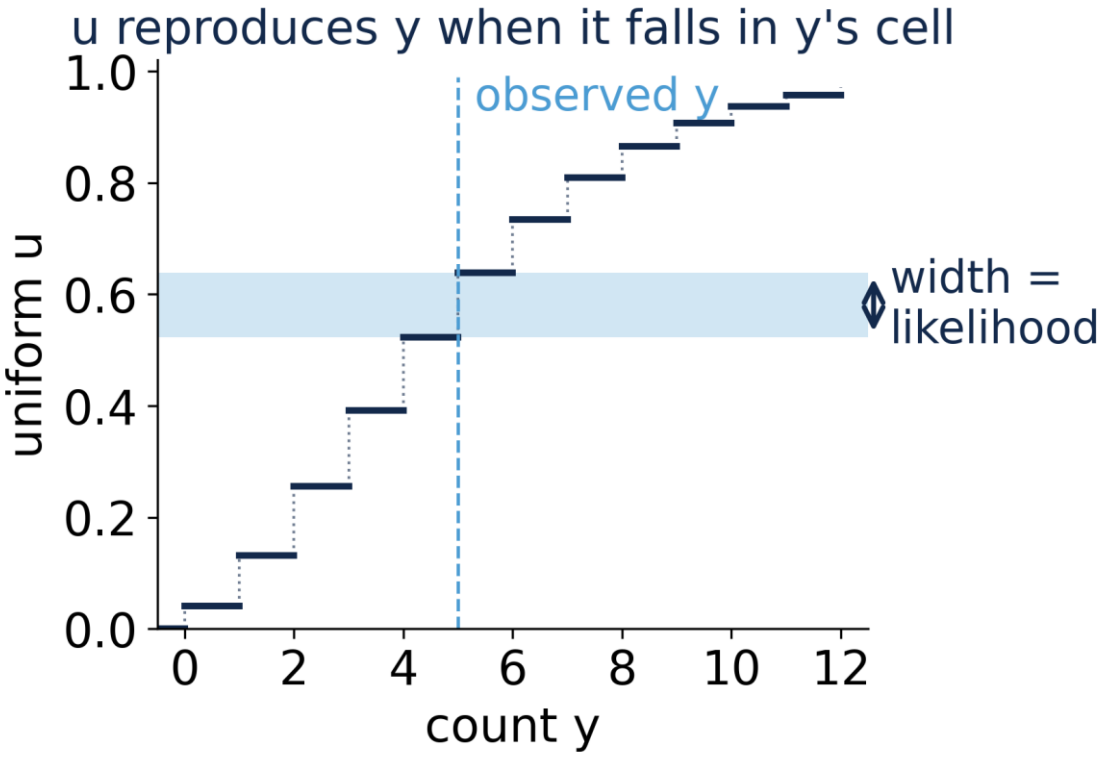
- **Proposition 3 (Block-Diagonal Jacobian):** The Jacobian decomposes into one block per movement bin, where each block depends only on that bin's own parameters $\theta_j = (\beta_j^{\mathrm{T}}, \nu_j)$.

$$J_C(\boldsymbol{\theta}) = \partial \mathrm{Y_C}/\partial \boldsymbol{\theta} = \mathrm{diag}(J_1, J_2, \ldots, J_{100})$$

$$J(\boldsymbol{\theta}, \mathbf{y_C}) = \sqrt{\det(\mathrm{n}^{-1} J_C(\boldsymbol{\theta})^{\mathrm{T}} J_C(\boldsymbol{\theta}))} = \prod_{j=1}^{100} \sqrt{\det\left(n^{-1} J_j^{\mathrm{T}} J_j\right)}$$

Discrete Rows: No Jacobian

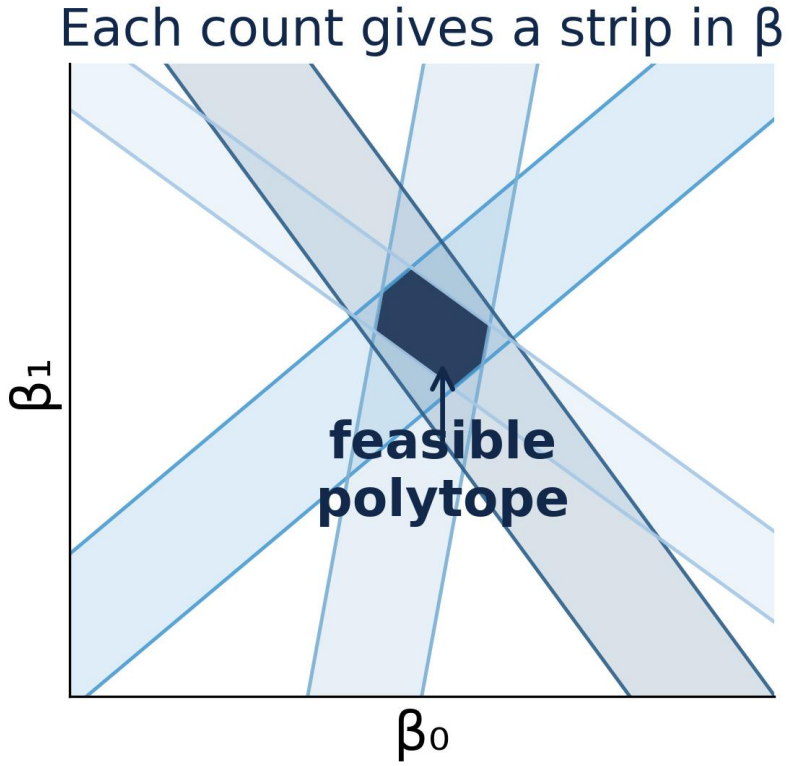
- **Fiducial Limit:** keep $\boldsymbol{\theta}$ with $\|G(\mathbf{U}, \boldsymbol{\theta}) - \mathbf{y}\| \leq \varepsilon$, let $\varepsilon \to 0$[3]
- **Integer responses:** for $\varepsilon < 1$, discrete rows must match exactly
- **Continuous rows:** pin $\boldsymbol{\theta}$ to an $O(\varepsilon)$ neighborhood
- **Discrete rows:** integrating over $\mathbf{U}_D$ gives the likelihood (right)



Jacobian: Normal rows only (≥ K + 1 per component, full column rank)
All-NB components still carry a selection correction (next slide)

Entirely Discrete Components

- **All-NB components:** no Normal rows, so no Jacobian: use the discrete construction[5]
- **Feasible parameters:** $\mathcal{Q}(\mathbf{y}, \mathbf{u}) = \{\boldsymbol{\theta}: G_D(\mathbf{u}, \boldsymbol{\theta}) = \mathbf{y}\}$
  - polytopes in β, one per feasible $log v$
- **Selection rule V:** $log v$ uniform, then β uniform on the polytope



Lemma 1: $r(\boldsymbol{\theta} \mid \mathbf{y}) \propto L(\mathbf{y} \mid \boldsymbol{\theta}) \mathbb{E}_{\boldsymbol{\theta}}[v_{\mathbf{U}}(\boldsymbol{\theta})]$

the expectation replaces the Jacobian; if it were constant, this is flat-prior Bayes

Bernstein–von Mises Theorem

- **Conditions (A)–(E):** bounded design, compact Θ, positive information

Theorem 5:

$$\|R_{\sqrt{n}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)|\mathbf{y}} - N(\mathcal{I}^{-1}\Delta_n, \mathcal{I}^{-1})\|_{TV} \to 0$$

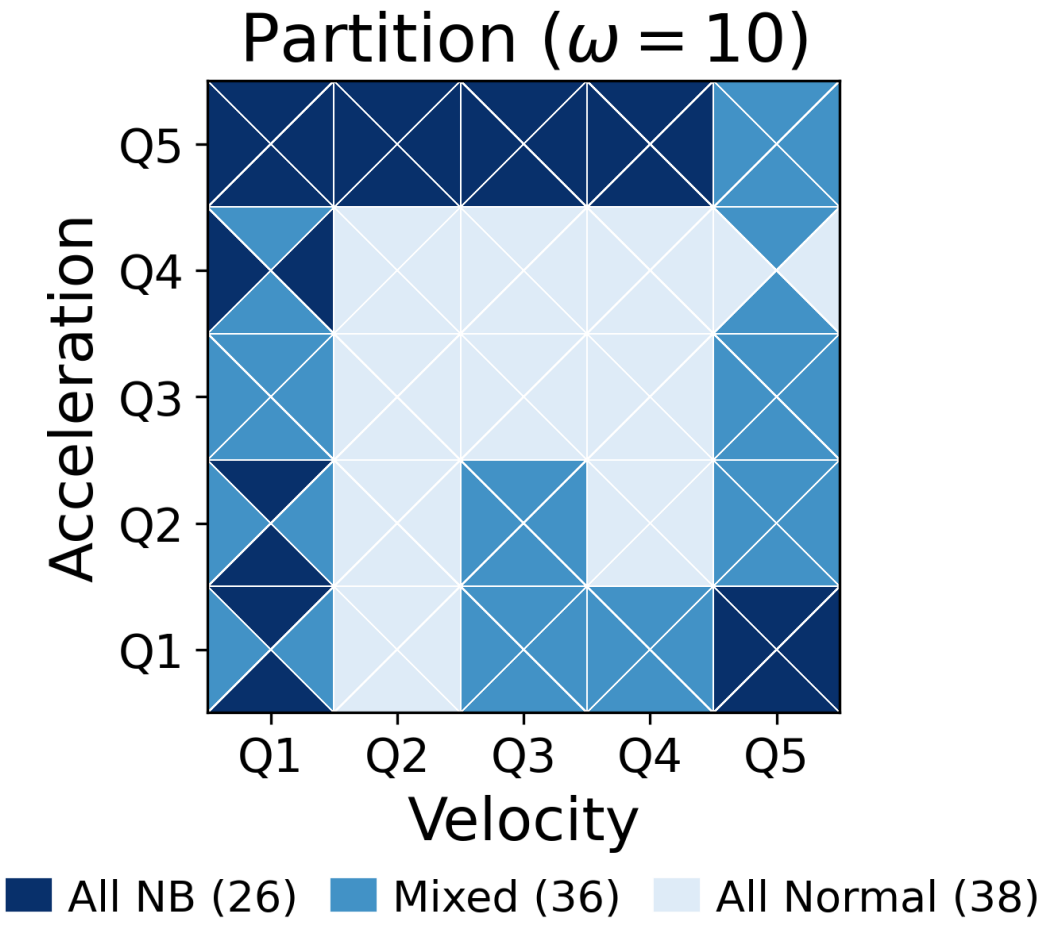
- **Proof strategy:** verify the five conditions of Borgert & Hannig[8]
  - Key Step: Jacobian $\to \pi(\boldsymbol{\theta}) = \prod\sqrt{detV_j} > 0$
- **Interpretation:** agrees with flat-prior Bayes to first order
- **Assumes:** oracle partition, correctly specified hybrid model, conditionally independent observations
  - not covered: the estimated partition, athlete and match dependence

Fitting the Model

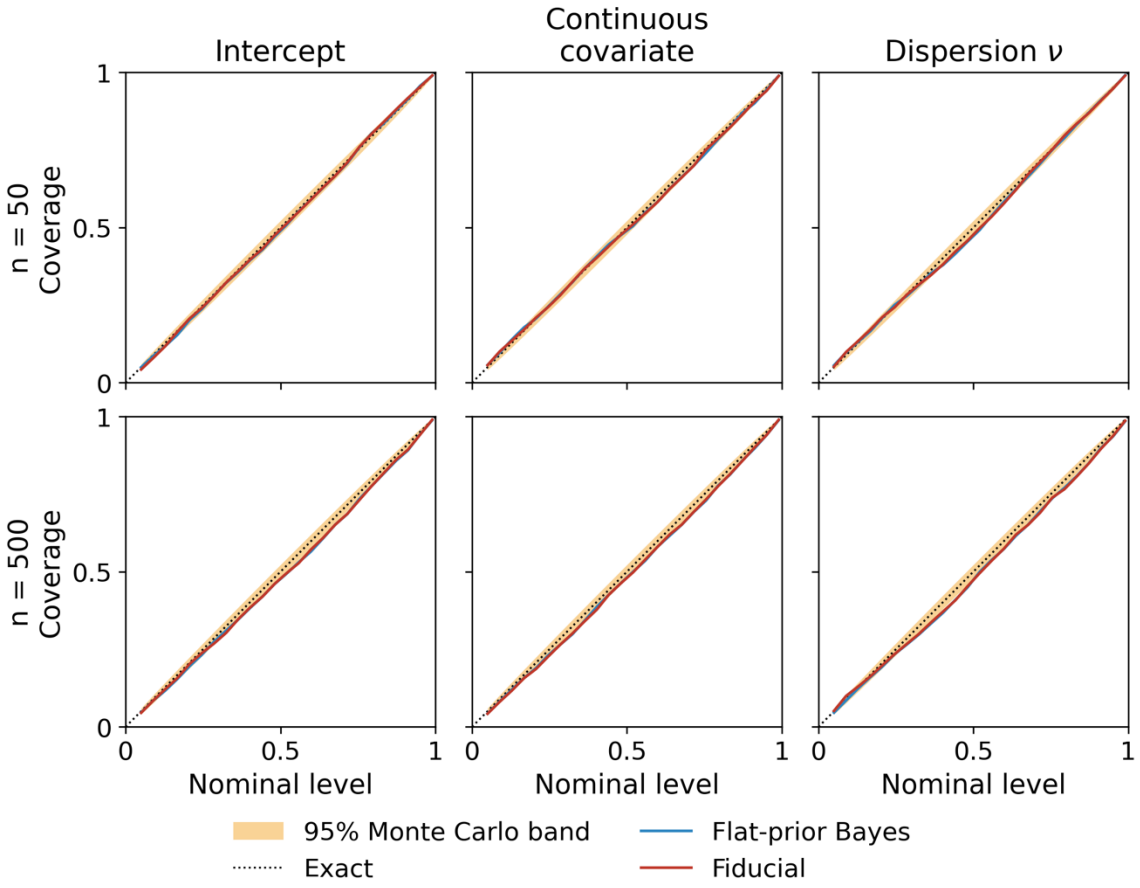
- **Lower-bound shape rule**: count i of bin j is Normal when

$$\hat{r}_{ij}^{L} = \hat{\mu}_{ij}\hat{v}_j e^{-1.645\hat{s}_j} \geq \omega = 10$$

  - $\hat{\mu}_{ij}$: pooled rate × exposure; $\hat{v}_j$: moment estimate; $\hat{s}_j$: athlete-bootstrap SD
- **Algorithm**: block-diagonal Jacobian, so each bin is fit independently
  - 74 active bins: Hamiltonian Monte Carlo, 4 × 4,000 draws, all $\hat{R} < 1.01$
  - 26 all-NB bins: exact discrete construction
- **Multiple testing**: Benjamini–Hochberg[4] FDR 0.05 per covariate

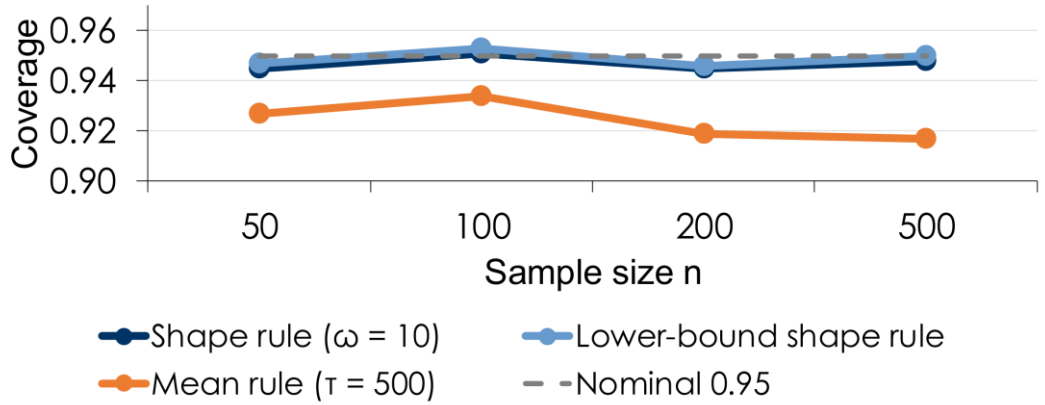


Simulation Results



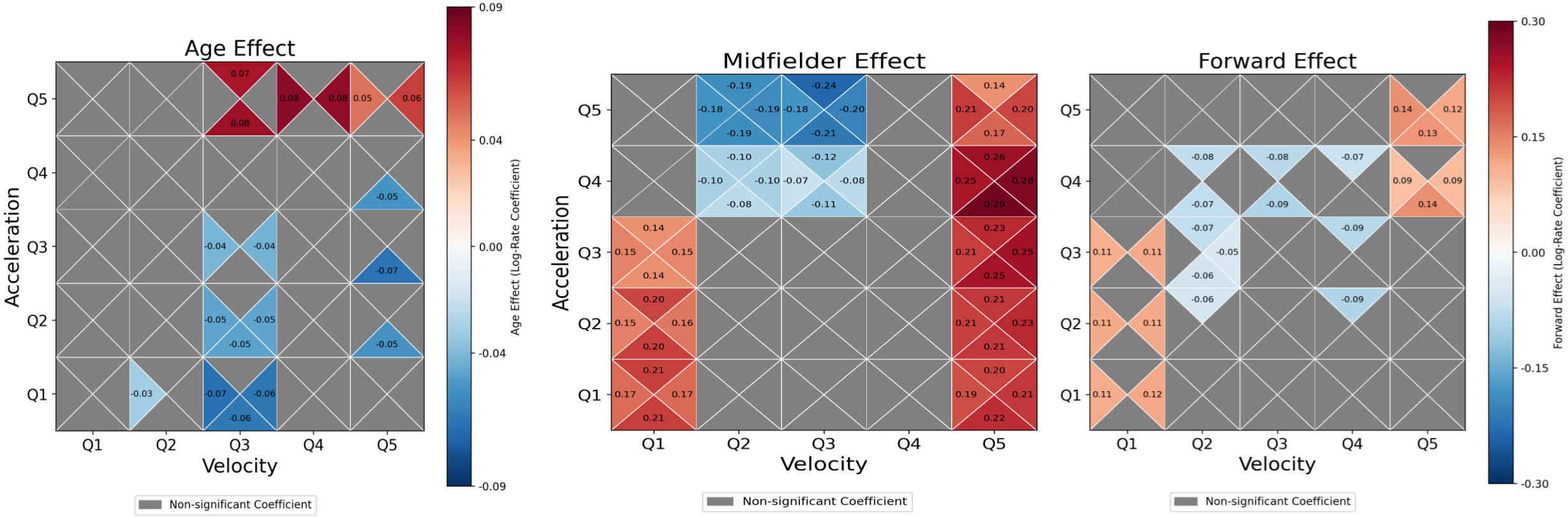
Coverage at every nominal level, borderline regime

- NB regression (intercept, continuous covariate, 3-level group, exposure); n = 50 to 500; 2,000 replications
- Average coverage of 95% intervals: **94.6%** (93–96%); calibrated at every level (left, borderline regime)



- Partition by shape (right): shape rules ≈ 0.95; mean rule ≈ 0.92; realized counts fall to 0.71

Results: Age and Position



- Declared (BH, FDR 0.05, working model): age **18** bins; forward **23** and midfielder **48** (vs. defender)
- Position effects reach $|\beta| = 0.29$ (midfielder, highest velocity); age effects stay below 0.09 per SD (2.1 yrs) and follow an acceleration gradient

Calibration Check

- **17 athletes, not 216 sessions:** permute position labels across athletes, or athletes' mean ages (keeping within-athlete deviations); refit all 100 bins, 100 times

Covariate	Bins declared	Median under permutation
Age	18	28.5
Forward	23	31
Midfielder	48	27.5

- Significance under the working model is not calibrated at the athlete level: the independent units are the 17 athletes
- The declared bins show where the fitted model places its largest, best-resolved contrasts: descriptive, not established

Bins declared: main analysis. The supplement's permutation table reruns the unpermuted data at its shorter chain settings, giving 19 for age.

Conclusion

Exact where it matters, approximate where it is safe.

Hybrid GFI

exact NB where the shape is small, Normal where it is large

block-diagonal Jacobian

Bernstein–von Mises

Simulations

near-nominal coverage

partition by estimated shape, not by the mean

Application

position effects exceed age effects

descriptive at the athlete level (17 athletes)

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The University of North Carolina at Chapel Hill

Thank you!
Questions?

Email: ketho@unc.edu

Website: https://kelanethomas.github.io/home/

Full Paper

quantile-cube Python Package Available on PyPI!

References

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